

Software Setup Required — Before Our Sessions Begin

We'll be working with Python programming, `.py` files, Jupyter Notebooks, VS Code, Git, and GitHub. Please complete **all** steps below before the first session.

1. Python

Used for: Writing and executing Python programs for Machine Learning, Data Science, and other practical sessions.

Download: <https://www.python.org/downloads/>

Recommended: install **Python 3.11 or 3.12** rather than the absolute latest release. Some data science libraries take a few months to catch up with brand-new Python versions, and this avoids install headaches on day one.

During installation, make sure to select:

- ☒ **Add Python to PATH**

Verify installation:

```
python --version
pip --version
```

Mac/Linux users: use `python3 --version` and `pip3 --version` instead.

Note: after installing Python, **restart VS Code (and your terminal)** before running the verify commands — otherwise PATH may not be updated and `python` won't be recognized yet.

2. VS Code

Used for: Writing and running `.py` files and working with Jupyter Notebooks.

Download: <https://code.visualstudio.com/>

Install these extensions in VS Code:

- Python — Microsoft
- Jupyter — Microsoft
- Pylance — Microsoft

3. Jupyter Notebook

Used for: Interactive Python coding, ML experiments, data analysis, visualizations, and step-by-step execution of code.

We'll use Jupyter Notebook inside VS Code.

Create a test file:

```
test.ipynb
```

Select a kernel: When you open `test.ipynb` for the first time, VS Code will ask you to select a kernel — choose the **Python interpreter** you just installed (top-right corner of the notebook, or via the "Select Kernel" prompt). If you skip this, your code won't run.

Run:

```
print("Jupyter is working!")
```

4. Git

Used for: Managing project versions and working with GitHub repositories using commands such as `clone`, `pull`, `add`, `commit`, and `push`.

Download:

- Windows: <https://git-scm.com/download/win>
- Mac: <https://git-scm.com/download/mac>
- Linux: usually pre-installed, or `sudo apt install git`

Verify:

```
git --version
```

One-time setup (required before your first commit):

```
git config --global user.name "Your Name"
git config --global user.email "your.email@example.com"
```

Use the same email as your GitHub account so your commits are linked to your profile.

5. GitHub

Used for: Storing and sharing our code/projects online and collaborating through Git repositories.

GitHub: <https://github.com/>

Your GitHub account should already be created and your email should be verified.

 **Important:** GitHub no longer accepts your account password when you `git push` over HTTPS. You'll need either:

- A **Personal Access Token** (Settings → Developer settings → Personal access tokens) used in place of your password the first time you push, or
- The built-in **Source Control panel in VS Code**, which handles login through your browser and avoids this issue entirely — recommended if you're new to Git.

Git Commands We'll Commonly Use

You don't need to memorize everything now, but you should be familiar with these commands.

Clone a repository — download an existing GitHub repository to your computer:

```
git clone <repository-url>
```

Check the current status — see which files have been modified or added:

```
git status
```

Pull the latest changes — get the latest changes from the `main` branch:

```
git pull origin main
```

Add changes — stage all changed/new files:

```
git add .
```

Commit changes — save your changes with a meaningful message:

```
git commit -m "Describe what you changed"
```

Example:

```
git commit -m "Added data preprocessing"
```

Push changes — upload your committed changes to GitHub:

```
git push origin main
```

View existing branches:

```
git branch
```

Switch to a branch:

```
git switch <branch-name>
```

Create and switch to a new branch:

```
git switch -c <branch-name>
```

Typical Git Workflow

When working on an existing project, the basic workflow will usually be:

1. `git pull origin main`

↓

2. Make your changes

↓

3. `git add .`

↓

4. `git commit -m "Describe what you changed"`

↓

5. `git push origin main`